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SCHOOL OF COMPUTER SCIENCE

Group Project Assignment: 30%

Agile Software Solution for Sustainable Development

Declaration of Originality:

We hereby declare that the report entitled "_____SPM-Warisan_____" submitted for the **module ITS 610204/ITS 64604:PRINCIPLES OF SOFTWARE ENGINEERING, Semester: February 2026** is entirely our own original work; no part of this report has been copied from other students, published works, or any other source without proper acknowledgment, and no unauthorized assistance (including any Al text generation tool) has been used in preparing this report, and we understand that any violation of this declaration may result in academic penalties.

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2 Abstract

This report presents the entire design and development of **"SPM-Warisan" app**, which is a second-handed book marketplace mobile application created for **ITS 610204/ITS 64604 Principles of Software Engineering** group project. This group project follows:

Sustainable Development Goal (SDG) 4: Quality Education by addressing the financial challenges students face when purchasing textbooks. For every year, students are required to buy textbooks that are used for only one academic year, which is expensive. This results in financial issues and also unnecessary waste. Due to these issues, **"SPM-Warisan"** provides a practical solution by allowing senior students to sell their used textbooks to junior students, which promotes affordability, accessibility, and resource sustainability. Other than that, **"SPM-Warisan"** is also a platform that allows senior students to donate their textbooks to junior students that are facing financial problems, who could not afford textbooks.

The project was developed using an Agile/Scrum methodology over a four-week period consisting of two sprints. During the process of development, scrum roles were assigned within the team to ensure structured collaboration, and sprint ceremonies such as planning, daily stand-ups, reviews, and retrospectives, which were important for maintaining progress and continuous improvement. Based on Software Development Life Cycle (SDLC) phases, including planning, analysis, design, prototype development, and testing were all implemented into the progress of this development. During this project, Trello was used to manage product and sprint backlogs. GitHub supported version control and documentation of the progress of the project.

In this project, key deliverables include UML Use Case and Class Diagrams, UI wireframes, burndown charts, a functional prototype, and structured user testing results. There are three user roles supported by the prototype, which are Junior (buyer), Senior (seller), and Admin with the features such as book search, book listing, book selection, in-app chat, notification system, and also bug reporting. User testing confirmed that the system meets its primary objectives and also provides an effective, student-centered solution to reduce textbook costs and support equitable access to education.

3 Project Introduction

3.1 Background

In Malaysia, access to affordable educational materials is a significant challenge for many students that are facing financial issues, especially in secondary schools and higher education institutions. Textbooks are often expensive, yet they are typically used for only one academic year and even one semester only before becoming redundant. This causes students to face recurring financial burdens while previously purchased books remain unused, contributing to unnecessary waste. This issue highlights both an economic and sustainability concern within the education system.

For example, some textbooks used in secondary schools may cost around RM50 or more, which is considered as a heavy expense for families having limited financial resources. Although school libraries provide some learning materials, the number of copies is limited and there are timelines for people to borrow. With this situation, some students who cannot afford these materials might struggle to access needed materials and this can affect their academic performance.

To solve this issue, the **"SPM-Warisan" app** is designed as a digital platform designed to support the exchange of second-hand textbooks among students at a reasonable price. The application acts as a marketplace where senior students who have completed a course can sell their used textbooks to junior students who are currently enrolled in the same subject. **"SPM-Warisan" app** creates a reliable and easily accessible platform that promotes resource reuse and lowers educational expenses by enabling direct transactions within the same institution. Other than that, the application encourages sustainability by reducing the waste generated from unused textbooks and promoting a culture of sharing within the student community.

3.2 SDG Justification

The **"SPM-Warisan" app** is designed to directly support UN Sustainable Development Goal 4 (Quality Education), which aims to ensure every student has access to high-quality education and opportunities to learn throughout their lives. Besides, this app also lowers the cost of textbooks, which allows students easier access to good and cheaper learning materials. This supports Target 4.1 (free, fair, and high-quality education) and Target 4.a (make learning spaces that are welcoming and improve them).

3.3 Target Users

In this project there are three primary user roles involved in this development. Junior users act as buyers searching for affordable textbooks. Students are able to browse book listings, filter books based on form level or subject, and also purchase textbooks at lower prices. Senior users function as sellers who upload and manage their book listings. Admin users oversee platform integrity by managing posts, monitoring reported issues, resolving reported bugs or inappropriate content, and ensuring that the platform. Each role has clearly defined responsibilities to ensure smooth system operation.

User Role	Description	Primary Role
Junior (Buyer)	Students looking to purchase and get textbooks based on their grade at affordable prices.	Find, view and purchase books by subjects and grades.
Senior (Seller)	Students who have completed courses and want to sell their textbooks.	Upload and manage book listings.
Admin	Platform administrator who is responsible for maintaining content quality system management.	Monitor posts, resolve issues, and maintain system integrity.

3.4 Project Objectives

The main objective of **"SPM-Warisan"** is to develop a functional mobile application prototype that demonstrates core marketplace features, including book search, listing upload, in-app chat, notifications, and admin management. These features allow users to interact efficiently within the platform and ensure smoothness in the transaction process.

The project also aims to apply Agile/Scrum methodology throughout the Software Development Life Cycle (SDLC), ensuring structured planning, iterative development, and continuous improvement. This ensures that the system design evolves based on feedback and usability considerations. Through this approach, the team seeks to produce a user-cantered, sustainable solution that contributes meaningfully to affordable education and also promotes the reuse of educational resources and contributes positively to improving access to education.

4 Agile Process and Trello Usage

To keep our project organized and on schedule, we utilize the Agile methodology with a Scrum-based approach. Trello served as our central hub for tracking tasks and ensuring every team member knew their responsibilities. Our board was organized into several columns to reflect our progress.

4.1 Product Backlog (Initial Planning)

The Product Backlog was our starting point where we listed all the high-level tasks required for the assignment. This helped us visualize the entire scope of the project before we started any design work.

Key Tasks:

Selecting SDG 4(Quality Education), setting up the GitHub Repository and brainstorming the initial system requirements.

Purpose:

This ensures that we didn’t miss any of the mandatory project deliverables from the start.

4.2 Sprint 1 (Foundational Design)

In the first sprint, we focused on the core logic of the app. This was about defining the user experience before building the actual prototype.

Key Tasks:

Developing User Stories for the Junior, Senior and Admin roles, and creating the first UI sketches for the Login and Home screens.

Purpose:

This allowed us to finalize how the students would interact with the app before we committed to the final design.

4.3 Sprint 2 (System Architecture and Final Prototype)

Sprint 2 was our most intensive phase, where we converted our initial ideas into formal software engineering diagrams and final prototypes.

Key Tasks:

Mapping out the Use Case and Class Diagrams and building the main app features.

Purpose:

To turn our requirements into structured logic and visual representation of how the app's features would be laid out.

4.4 In Progress

This column was used to track what everyone was working on during the week to avoid confusion.

It allowed us to:

i. Track our progress
ii. Avoid duplicate work
iii. Ensure tasks are completed according to the sprint plan.

4.5 Test / Review (Control Quality)

Since our project is a UI mockup and not a coded application, our testing phase focused on internal quality checks. Once we finished the prototype, we moved them here to be reviewed or tested by us.

Self-Evaluation:

We checked the Figma prototype screens against our original requirements to make sure we didn't miss any important features, for example the "Subject" selection for the book filtering system.

Logic Check:

We reviewed the navigation flow of our mockups to ensure the transition between pages felt logical and followed a student's perspective.

Purpose:

The main point of this phase was to double-check our work before turning it to the final prototype. Since we didn't have a working app to test for bugs, we tested our ideas instead.

4.6 Done

This was our final archive for every small task. As soon as a diagram, a user story, or a screen was reviewed and finished, we moved the card here. It was the best way to see what was completed each week.

Purpose:

The goal was to keep the project moving and make sure no one was confused about what was left to do.

5.0 SDLC Phases

5.1 Planning / Requirements

Why do we create SPM-Warisan?

Our project focuses on SDG 4: Quality Education, specifically addressing the challenges faced in Malaysian education in Sekolah Menengah Kebangsaan (SMK) schools. The core problem is the financial burden of reference books and exercise books on students and families. For example, a new Biology reference book can cost at least RM 40, which is a significant expense. While libraries exist, they have limited copies and strict borrowing periods. We chose to create a peer-to-peer book exchange system because it directly tackles this financial gap by enabling students to share books for free or at a minimal cost. This is both a sustainable and community-driven solution than simply fundraising to buy new books, as it empowers students to help each other.

Personas

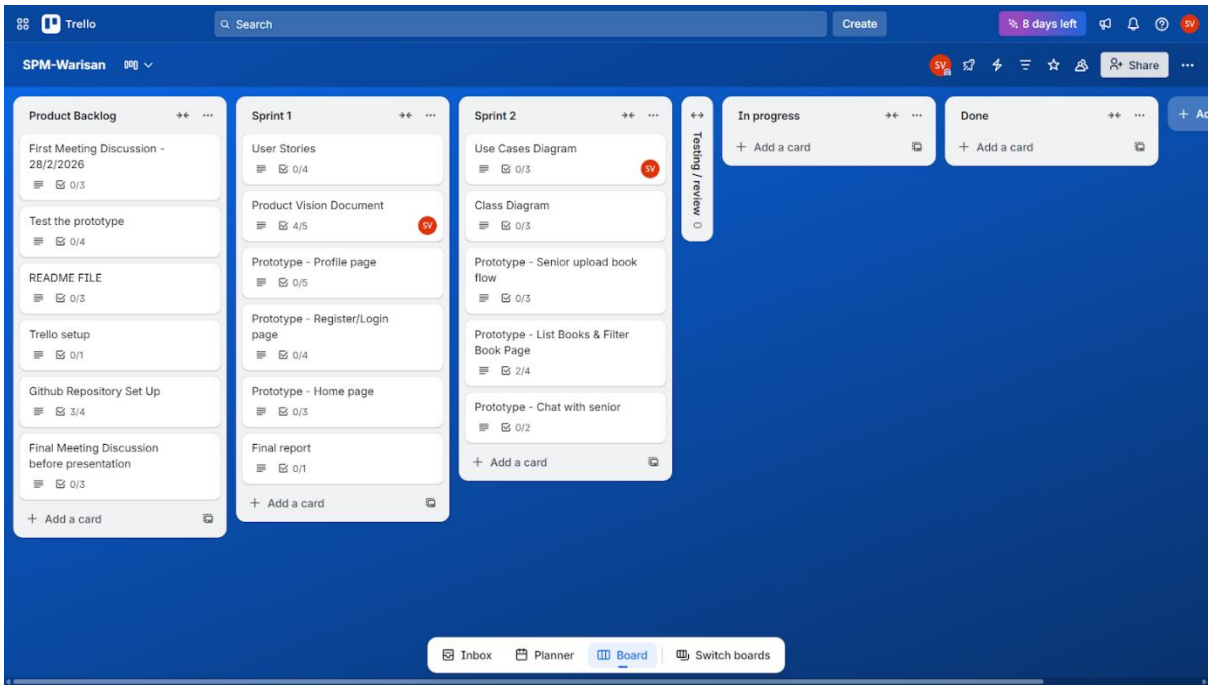
Junior (The Seeker) : Ahmad is a Form 2 student. He is very diligent and hardworking but comes from a low-income family. His pain is that he cannot afford the expensive reference book that his teachers recommend, putting him at a disadvantage compared to his peers. He needs an easy way to find books without spending too much money.

Senior (The Giver) : Siti is a Form 5 student who just finished her SPM. She has a collection of well-kept reference books from all her years of studying. Her pain is that these books take up a lot of space in her room and she wants to pass them to someone who needs them, but all her friends graduated as well.

Admin (The Guardian) : Mr Tan is a senior assistant teacher. His pain is managing the physical school library and dealing with lost or damaged books. He worries about the appropriateness of student-to-student interactions and wants a system that is safe, monitored and free of inappropriate content.

The Product Backlog

To manage the project scope from the beginning, we created an initial Product Backlog using Trello.



Explanation of our Trello Board

Product Backlog (Basic Planning)

This is where we put all the basic planning tasks at the very beginning. It includes things like our first meeting, setting up Github and writing a README file. It helped us to see the big picture of what needed to be done for the whole project.

Sprint 1(Early and easier project work)

Here, we focused on early parts of the project that were a bit easier, like writing our User Stories and creating the first few easier prototype screens (login,home page). Overall, this is where we built the foundation of the ideas.

Sprint 2 (The harder work)

Sprint 2 was much more complex than Sprint 2. This is where we did the harder parts like mapping out the Use Cases and Class Diagrams. We also had to design the main feature for our prototype like chat with the senior system and senior upload book flow.

Testing/Review

We added this column for when we finished our prototype screens. It's our way to make sure the design looks right and matches what we planned before.

In Progress

This is where we store the tasks we are working on right now. It helped me as a leader to see who is currently busy, so we don't double up on the same work.

Done

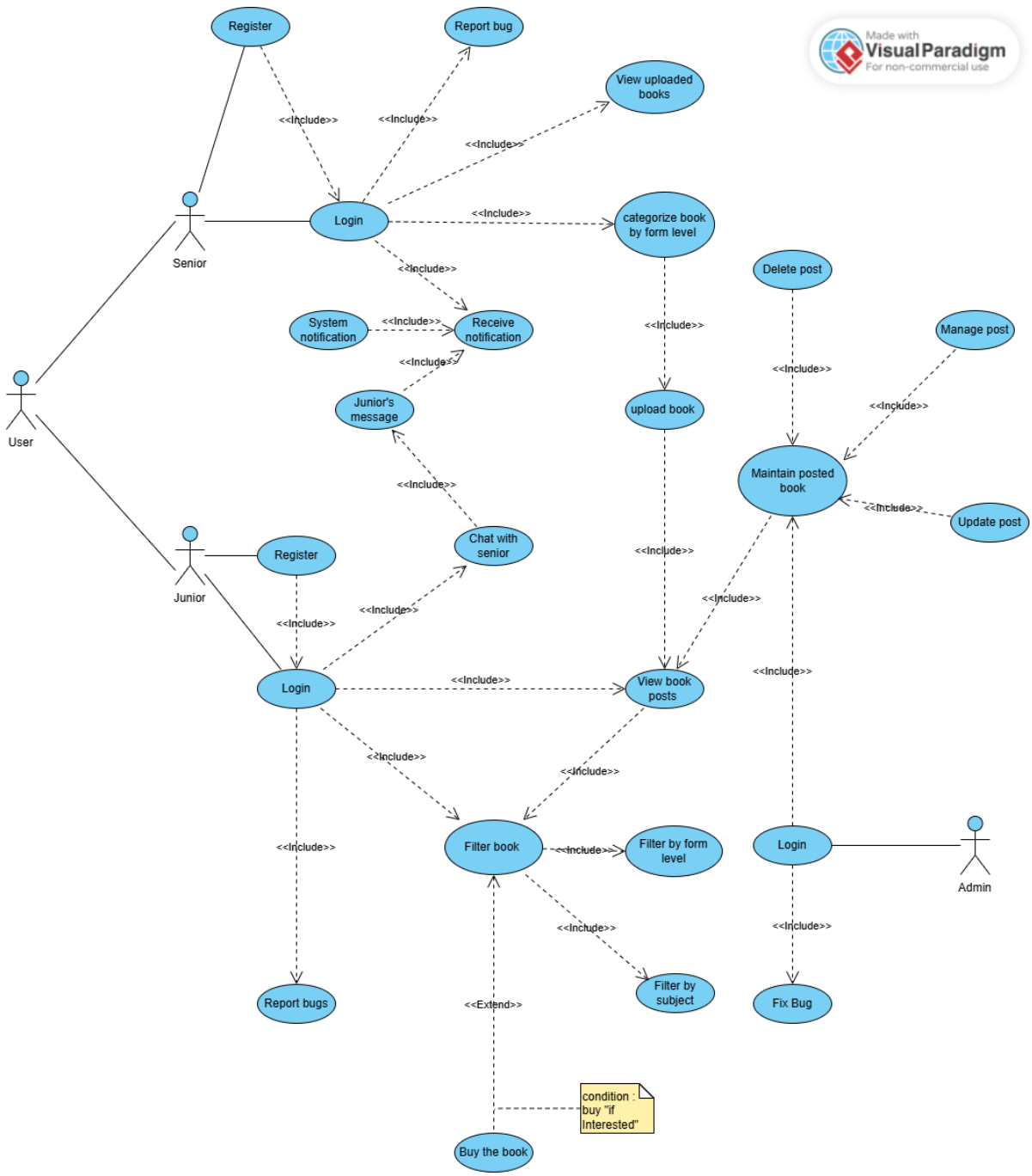
Once the task is 100% completed and checked, we move it here.

5.2 Analysis

Use Case Diagram

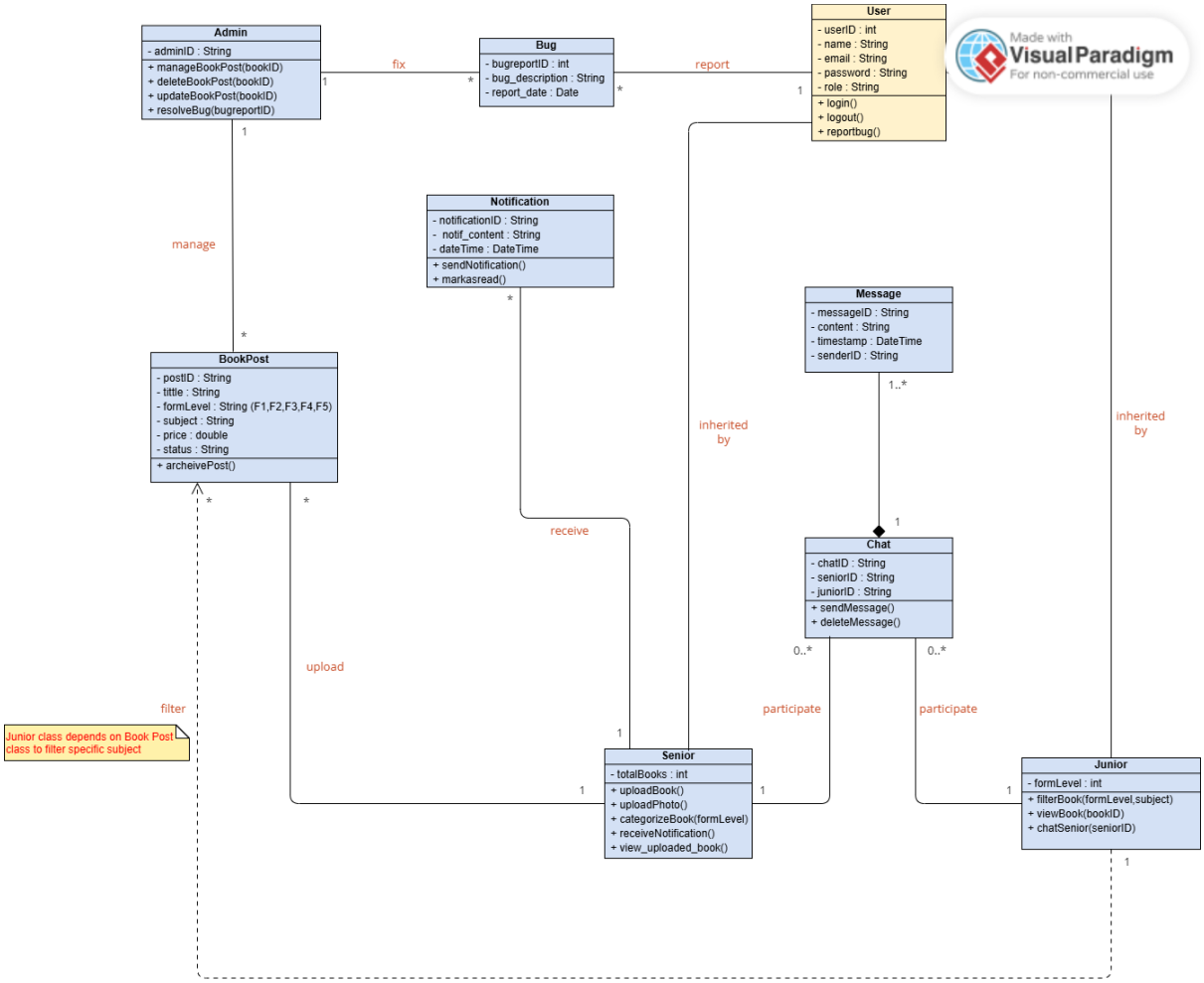
The Use Case Diagram below models the functional requirements of our Book Exchange system called "SPM Warisan ". It shows the interactions between the actors and the system.

The actors include the User, which is separated into Senior and Junior, there is also the admin which is another actor. Once a user is logged in, there are several actions that are included in the "Login" use case, which includes "Report Bug ", "View uploaded books ", "Categorize book by Form level "and so on. An extended function is shown between "Filter Book "and "buy the book "with the condition of "if interested "which allows the users to purchase the book.


Register
Report bug
View uploaded books
<<Include>>
<<Include>>
<<Include>>
Senior
Login
<<Include>>
categorize book by form level
<<Include>>
Delete post
System notification
<<Include>>
Receive notification
<<Include>>
Manage post
<<Include>>
<<Include>>
User
Junior's message
upload book
<<Include>>
Maintain posted book
<<Include>>
Update post
<<Include>>
Chat with senior
<<Include>>
Junior
Register
<<Include>>
<<Include>>
<<Include>>
Login
<<Include>>
View book posts
<<Include>>
<<Include>>
<<Include>>
<<Include>>
Filter book
<<Include>>
Filter by form level
Login
Admin
<<Include>>
<<Include>>
<<Include>>
Report bugs
<<Extend>>
Filter by subject
Fix Bug
condition : buy "if Interested"
Buy the book

Class Diagram

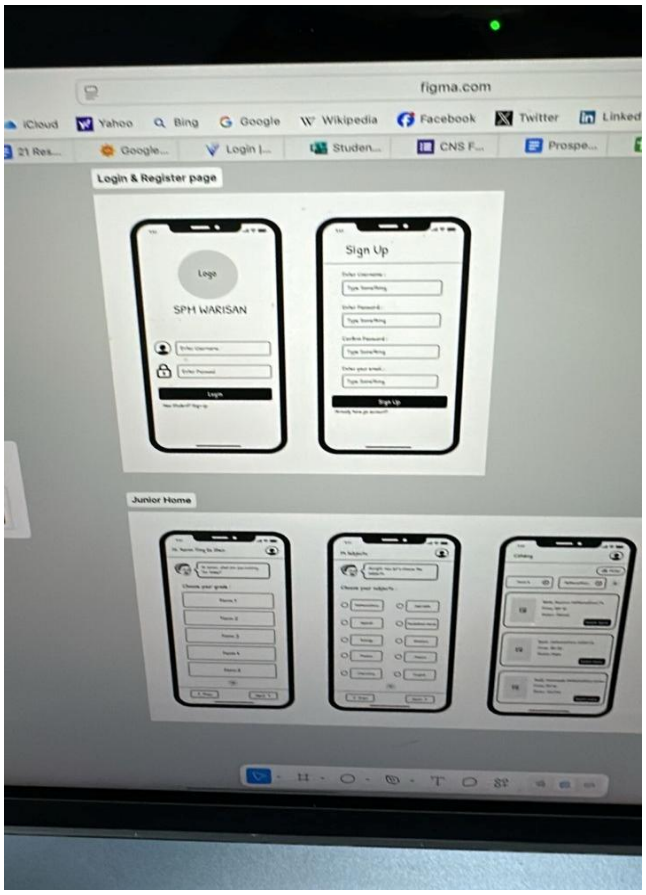
The Class Diagram outlines the static structure of our system, detailing the data entities and their relationships. For example, the “User” class is abstract, with concrete subclasses “Senior” and “Junior” which both have several distinct attributes like “totalBooks” and “formLevel”, each being only available in the separate user. The class diagram also shows multiplicity, A “Senior” can have one or more “BookPost” but each “BookPost” can only belong to one “Senior”. The “Admin” class is also directly connected to the “Bug” class illustrating the “Admin” can manage multiple bug reports.



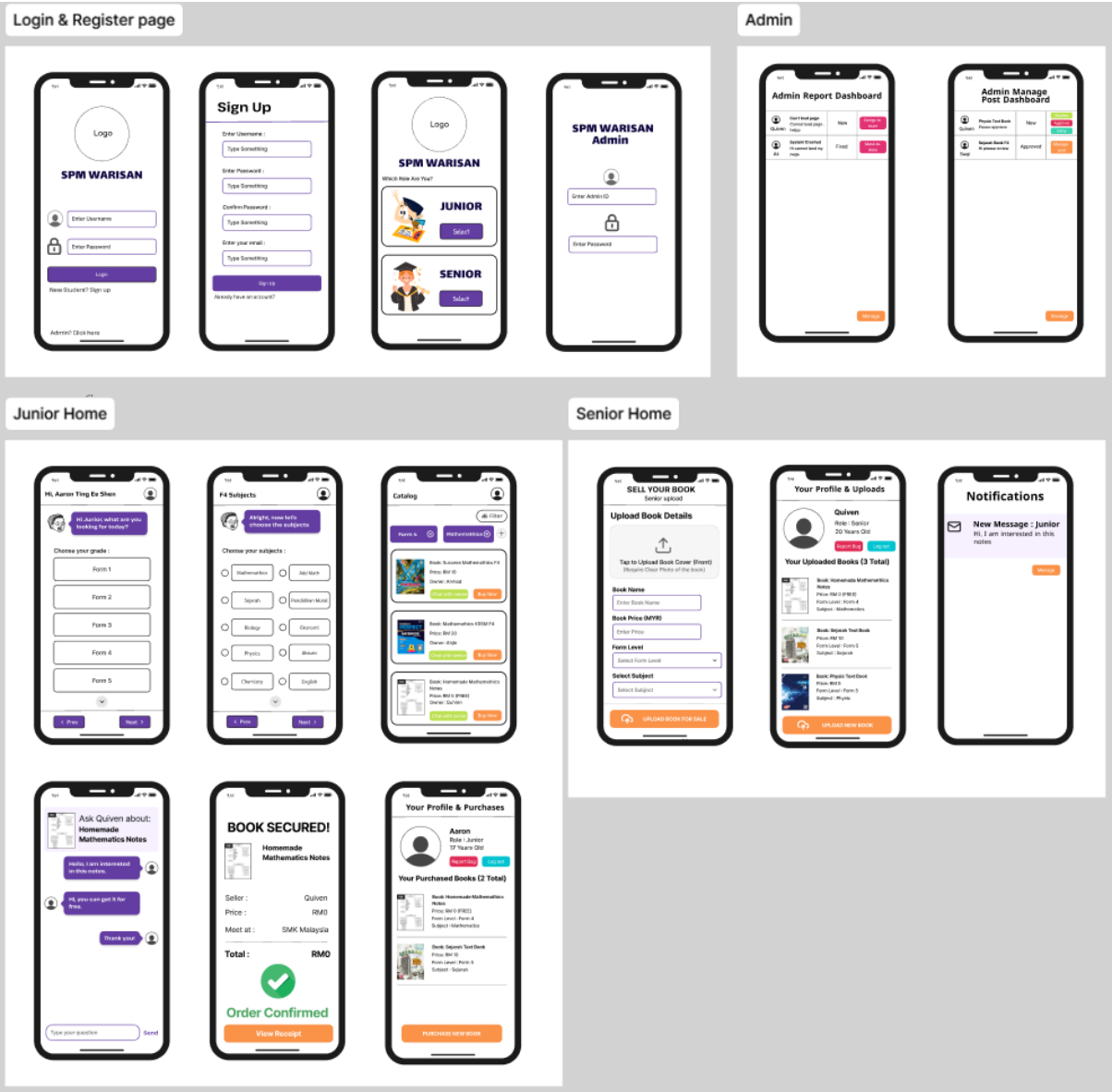
5.3 Design

From wireframes to mockups

Our design process began with low-fidelity wireframes to quickly iterate on layout and information architecture, without being distracted by colours or fonts.



These evolved into high-fidelity mockups created in Figma, which defined the final look and feel, including typography, colour scheme and interactive elements.



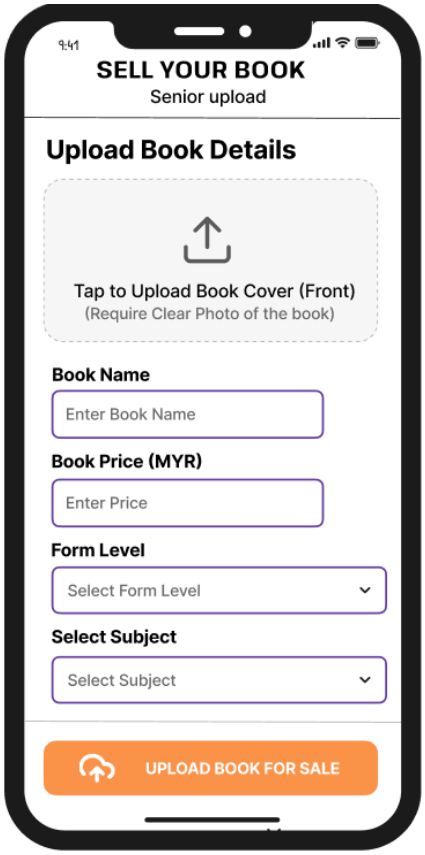
Design justification

A key design was placing prominent “form 1-5” and “subjects” category buttons directly on the home screen. This decision was made to prioritise the primary user goal for Junior students which is finding books that are relevant to their current studies. By reducing the number of clicks to reach relevant content, we aim to create a faster, more intuitive experience for our core user group.

Consistency (UI & Class Diagram)

We ensured strict consistency between our logical Class Diagram and our UI. For instance, the BookPost class in our diagram includes attributes like “title”, “formLevel” and “price”.

This is directly reflected in our “Upload Book” screen, which contains input fields for “Book Name” and “Book Price” , a dropdown menu for “Form Level” and “Select Subject”.



5.4 Prototype Development

Phase 1 (Building the basic flow):

After we did our Use Cases and Class diagrams, we created the main entry points of the app including Login and Register screens, and the Home page. We wanted to make sure that the first screen the users see is clean and easy to understand.

Phase 2 (Adding main features):

For senior students, we build the flow to upload a book photo, set a price and add the book name. For the junior students, we designed the Catalog where they can filter books by subjects as well as Form level (like Form 3, Mathematics) and purchase them for a specific price in RM.

Visual Design and Layout

We chose to use Figma to design our app. We used the colours and icon that fit perfectly for an education app because we wanted to make sure the app looks professional and clean.

5.5 Testing / Feedback

How we checked our design

Since our prototype is just a design that only shows the screens as well as ideas and it is not a real app with a backend, we did a self-evaluation. We went through every single screen we designed in Figma to make sure the flow was easy to use for a student.

What was fixed during this Phase?

During our self-check, we found a few things that need to be better:

The Problem:

At first, we wanted users to use the **Search** (type in box) system to find a book.

The Realization:

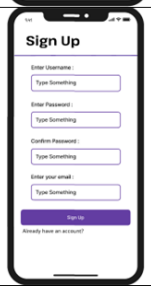
I then realized this app is just for Form 1 to 5 students with specific subjects. Typing a search might be slow or confusing if they misspell a word.

The Fix:

We changed the design to use a **Filter system** instead.Now students can just click their Form level and their Subject. This is much easier and faster for them to find exactly what they need.

Manual Test Cases Table

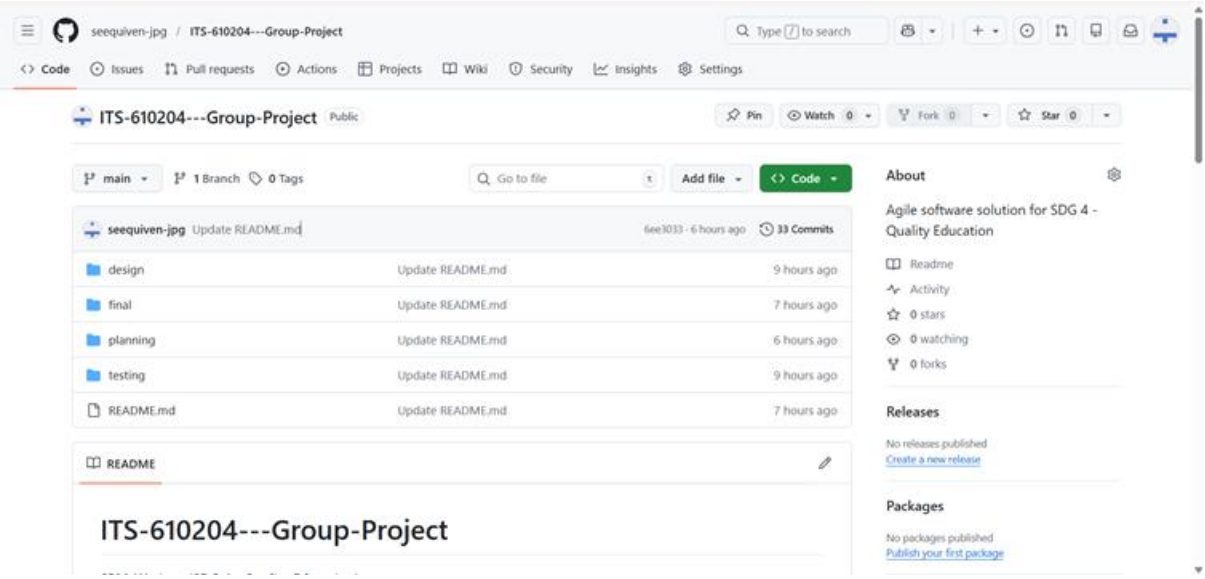
We used a simple test cases table to record these checks, for example the diagram shown below. It contained Test Cases ID,Test Case Description, Prerequisites, Test steps, Test Data, Expected Results , Actual Result, Status ,Created By Who, Date By Creation, Executed By Who, and Image Evidence. We also verified the Login, Filtering and Chat screens all look right and work in a logical order. This proves that our design is ready and easy for humans to use.

Test Case ID	Test Case Description	Prerequisites	Test steps	Test data	Expected Result	Actual Result	Status	Created by	Date by creation	Executed By	Execution date	Image
TC001	Check the Login page layout	Open the app and direct us to the first screen	1.Look for the SPM Warisan name 2.Check for Username and Password boxes 3.Find the purple Login button	No data needed	User should see the app name, two input boxes and a login button	All items are showed clearly on the first screen	Pass	See Qui Ven	5/3/2026	See Qui Ven	5/3/20206	
TC002	Check the sign up pag	Go to the Sign Up screen	1.Look for the Username and Email boxes. 2.Check for "password" and "comfirm password" boxes	Enter the usernam e, email and passwor d	There should be 4 boxes for user to fill in their details	The drawing has all 4 boxes needed for a new user	Pass	See Qui Ven	5/3/2026	See Qui Ven	5/3/2026	

6.Github Evidence

6.1 Repository Setup

Github Repository is where we keep all our work. We set up a shared Github repository at the very beginning of the project so that we have a safe place to store our files and track our progress. Since we wanted to maintain a high standard for our work, we agreed that the team would send their finished tasks to me first, and I acted as the final reviewer to check for any mistakes before officially uploading the files to the repository. This helped us keep the project organized and ensured that version we submitted was the best one possible.



6.2 README File

Our README file is basically the introduction to our whole project. We wanted it to be clear and helpful for anyone looking at our repository for the first time. Inside, we included our project title, our focus on this chosen topic, and a list of our team members and what each of us is responsible for. We also added a short summary of the SDLC phases we followed so it's easy to see the logic behind how we built our app idea. It's a simple way to show the big picture of what we were trying to achieve with SPM-Warisan. Below are out README file screenshots:

README

ITS-610204---Group-Project

SPM-Warisan (SDG 4 - Quality Education):

The mission of SPM-Warisan is to make learning materials affordable for SMK (Sekolah Menengah Kebangsaan, which means National Secondary School) students, with a primary focus on B40 families. In Malaysia, thousands of study materials such as textbooks, reference books and student's own notes are discarded after SPM, our platform helps transform this waste into another valuable knowledge for the next generation.

Team Members & Roles

See Qui Ven - Project Leader and Scrum Master

Aaron Ting Ee Shen - Prototype Designer

Micheal - Class Diagram Creator

Joe - Report Writer

Jason - Use Cases Diagram Creator

What have we done so far

- Product Vision Document ✅
- Github setup ✅
- Trello setup ✅

No releases published
Create a new release

Packages

No packages published
Publish your first package

Contributors 1

seequiven-jpg

README

Project Description

SPM-Warisan is a software solution designed for the Malaysian SMK community. After SPM examination, thousands of study materials are often thrown away or left unused. Our platform creates a "relay" system where graduates can either donate or sell their study materials at very low cost to lower form students, especially for those who can't afford these learning materials.

Who it help?

Seniors - Able to earn a small amount of money back, or gain charity points by donating.

Juniors - Students, especially from B40 families can reduce the burden and continue their learning journey.

Environment - Help reduce paper waste.

Overview of SDLC Work

We follow Agile Scrum:

1.Planning - First, we discussed and brainstormed which topic is suitable and meaningful for us as a student.After that we decided to choose SDG 4 - Quality Education.During this phase, we wrote down our User Stories to clearly define what Junior, Senior and Admin need in the app.After that, we also set up trello board to keep track of our progress so that we wouldn't miss the week 6 deadline.

2.Design - Once we comfirmed with our goal, we started to turn our ideas into clear plans.Our analysts started to map out the Use Cases and Class Diagrams to show how the system should function and how different users should interact with it. After the diagrams were done, we then discussed how our prototype should look like to meet the actors requirement.

3.Building the prototype - Using the sketches, our prototype designer used Figma to build the main screens for our idea. We didn't make the whole system work (backend or store real data) because this project is time-limited, but we made it visual walkthrough. This screens show exactly how a user would move through the app. For example, starting from a Junior searching for a book until they reach the "Book Secured" page. Hence, this prototype shows our full idea even though it is just screens.

4.Testing and feedback - Since our prototype is a static design as well as idea and not functionable app, we perform a self-evaluation for our testing phase. We acted as the users and went through every screen to make sure the design followed our User Stories. Not only that, we checked all the important buttons (like Buy Now or Chat with senior) were easy to find and that a flow from one screen to another made sense. This manual check helped us to make sure the layout and design were clean.

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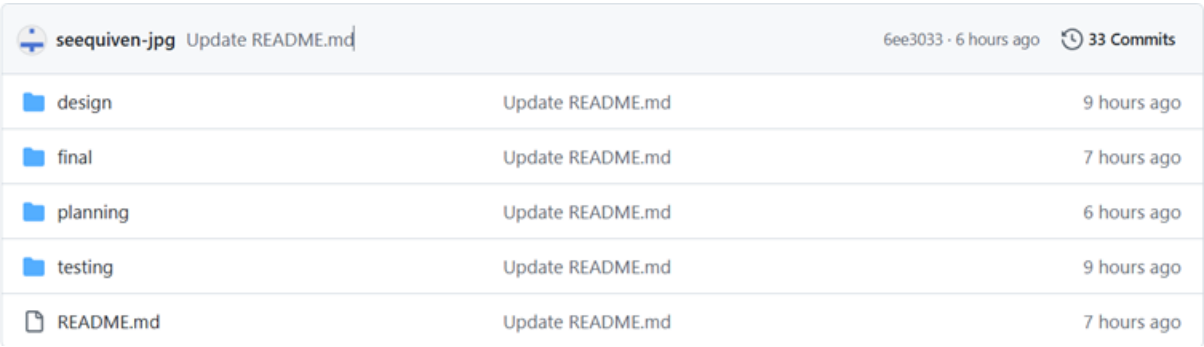
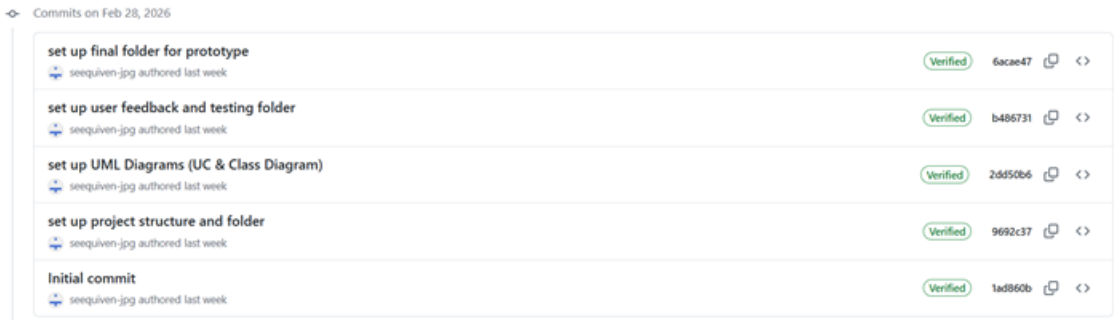
6.3 Folder Organization

To meet the organization standards of the assignment, the repository is structured into specific folders that represent each stage of the SDLC. In the **planning** folder, we kept our Trello screenshots and user stories. The **design** folder contains our UML work like Use Case and Class Diagrams. We also used a **testing** folder for our test cases and **final** folder for the finished prototype and the report. This structure made it simple to manage our files and show the progression of the project.



6.4 Commit History

We updated the repository regularly as tasks were finished to show a clear timeline of our work. Even though my name appears for the uploads since I was doing the final checks, these commits represent the collective effort of the whole team over the last few weeks.



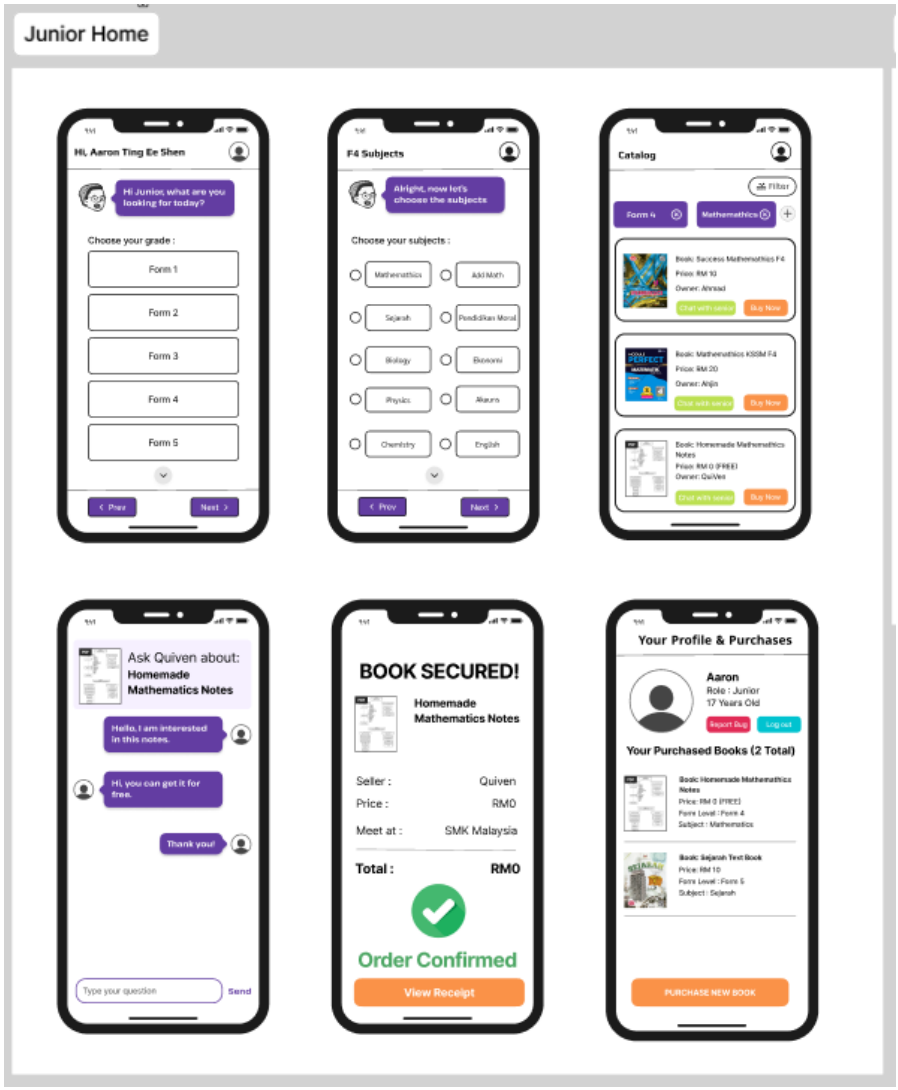
7 Prototype Showcase

7.1 Our Design Approach

For our prototype, we really wanted to focus on how a student uses the “SPM-Warisan” app. Since our goal is SDG 4, which is Quality Education, we aim to design the screens to be as clean as possible so that the users don’t get overwhelmed. These mockups aren’t a working system yet, but they show our vision for the app and how we translated our UML Diagrams (Use Cases and Class Diagrams) and User Stories into a visual layout.

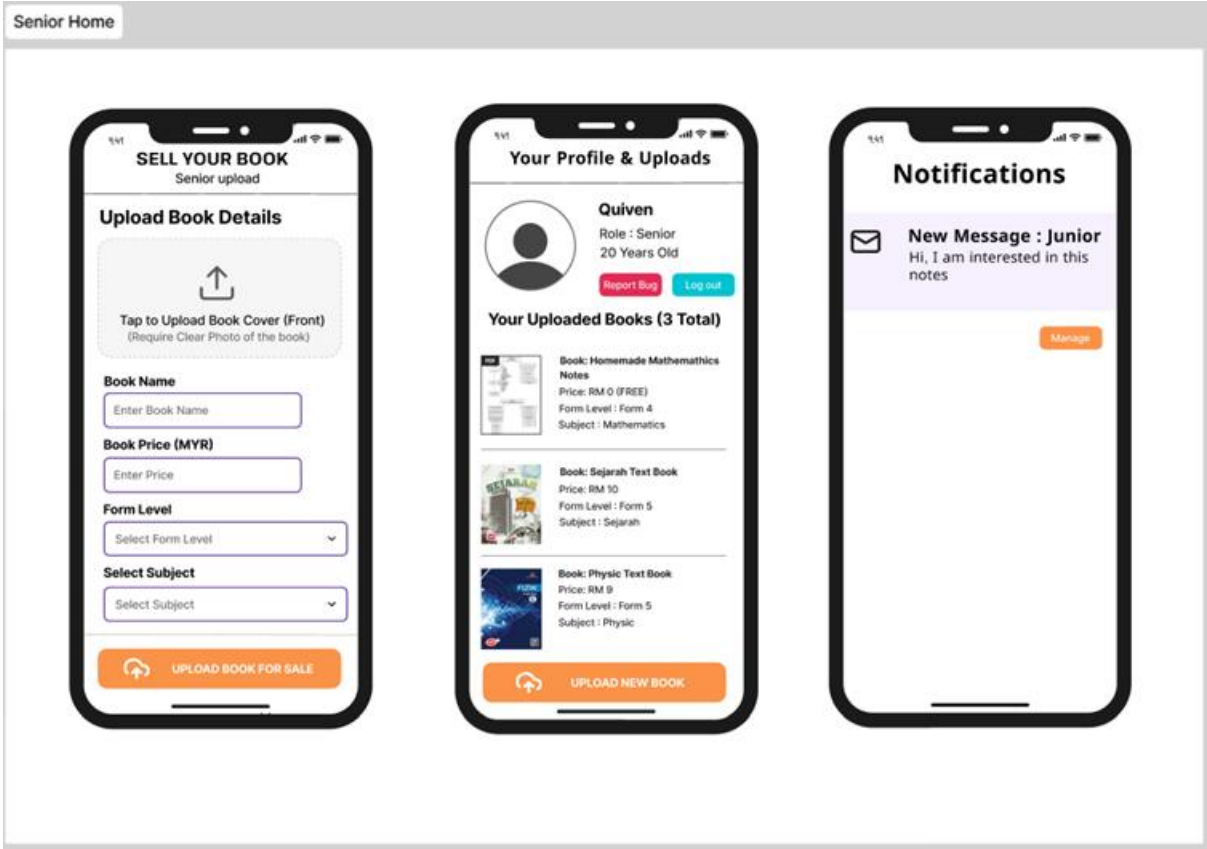
7.2 Junior Student View

We started by designing the Junior student's perspective. We created a Home Page where a student can immediately pick their form level, like Form 1 or Form 5, to see the right books. We also included a mockup of a chat screen because the whole idea is for Junior to message the Senior and ask about the book. We kept the colors simple so that the app feels easy to use for any student.



7.3 Senior Student View

For the seniors, we focused on the Upload process. We designed a screen where a senior can take photos of their old books, type in a title, and select the correct subject category. Our goal here was to make it as fast as possible for them to post their materials. We also made a basic profile page so they can keep track of what they've uploaded with the community. We wanted the design to encourage them to pass down their old study materials instead of just throwing them away.



7.4 Visual Logic and Consistency

Even though these are just static screens, we made sure the design stays consistent across the whole app. We used the same navigation bar and font styles on every page, so it feels like a real application. This showcase is basically the final result of all our planning, it shows that our idea works as visual UI design, even if the buttons aren’t clickable yet.

8 Reflection and What We Learned

8.1 Figuring Out How to Start

When we first started this project, we were honestly quite lost because we had no idea where to start. It felt like a mountain of work. But I then remembered something from the lecture slides about a **Modular** approach, which basically means breaking a giant project into tiny, little pieces. We decided to try that, and it changed everything for us. Instead of worrying about the whole report, we just focused on one small task at a time. It felt so good to see out progress day b y day, and it taught us that even the hardest project is easy to finish if you just keep moving.

8.2 Finding a Better Way (Filter vs Search)

One of the coolest moments was during our testing phase. At first, we were so sure that a search bar was the best way for students to find books. But when we sat down and looked at our screens, we realized we were wrong. Typing might be very slow for a student, and they might not even know what to search for. If they make a typo or spell a subject name wrong, they won't be able to find anything at all. Hence, we found that a simple Filter system, where they can just tap their Form level, was much easier and faster. We feel proud of that moment.

8.3 What I Learned as a Leader

This project taught me a lot about myself. I realized that I'm the type of person who wants to finish everything immediately. I usually finish my tasks in one day because I really hate waiting until the last minute. Because I work so fast, I noticed that I didn't have much patience when my teammates took longer to finish their parts. I know this probably puts a lot of pressure on them. It made me realize that being a leader isn't just about being fast, it's about being patient and understanding that everyone has their own speed. In the future, I want to be better at guiding my team without making them feel rushed.

8.4 Final Thoughts

Even with some stressful moments, we grew a lot as a team. We went from being confused to feeling really proud of what we built. We learned how to talk to each other better and how to fix our own mistakes. We are happy with how "SPM-Warisan" turned out, and we are glad we worked on something that could actually help other students, especially those who can't afford expensive study materials.

9. Conclusion

To wrap everything up, this project was a huge lesson for us. We followed the SDLC steps from start to end, and it showed us how much planning goes into an app. Even though our prototype is just a design and not a real app yet, we are proud of how we stayed organized and worked together to reach the finish line on time.

Github link :

https://github.com/seequiven-jpg/ITS-610204---Group-Project

Trello link :

https://trello.com/b/JVYUY4NE/spm-warisan

Figma link:

https://www.figma.com/design/lAhuYstWrUnkGLJUFnyvLK/Paper-Wireframe-Kit--Community-?node-id=5190-533&p=f&t=nuhSaH6bt5uR2Ucc-0

Other source:

Everything is in the github